Temi Otun

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Education

University of Alberta Sep. 2022 – Present
Bachelor of Science, Computing Science – Artificial Intelligence Specialization, Minor in Business *Edmonton, AB*

- **Awards:** Jason Lang Scholarship, Dean's List

Experience

ML Data Engineering Intern May 2026 – Aug. 2026
Royal Bank of Canada (RBC) – Amplify Program *Vancouver, BC*

- Developed a **patent-pending fraud detection system** combining ML classification with LLM analysis, achieving **94% recall** and an **85% F1-score**, projected to mitigate **$18M in annual fraud losses**
- Engineered a cascaded NLP pipeline processing **23M+ annual unstructured text records** using semantic embeddings and multi-stage classification, reducing operational costs by **65%** (**$1.5M in projected annual savings**)
- Built FastAPI REST services with **PostgreSQL**-backed asynchronous job tracking, load-tested to **1,000** requests per second through concurrent execution; secured with **OAuth2/OIDC**, **Keycloak**, and **JWT**-based **role access control**

Machine Learning Research Intern Jan. 2025 – Aug. 2025
The Metabolomics Innovation Centre - SOIL-HUB *Edmonton, AB*

- Built Python and MySQL pipelines integrating local weather observations with global climate indices across **82 Alberta counties** and **63 years** of data to support agricultural weather forecasting
- Developed **CatBoost models** for **3 county clusters** and **12 forecast horizons**, using **132 lagged features** to capture seasonal patterns and regional weather differences
- Achieved **75.4% temperature-category accuracy**, exceeding climatology by **42.1 percentage points**, evaluated across **259,776 predictions** spanning **22** test years

Machine Learning Research Assistant Sept. 2024 – Dec. 2024
University of Alberta *Edmonton, AB*

- Developed a three-class cognitive impairment detection system using Whisper-derived speech embeddings and Random Forest and XGBoost classifiers to distinguish healthy controls, mild cognitive impairment, and dementia
- Built survival-analysis and clinical prediction pipelines on MIMIC-IV, processing **59M+ patient-event records** across a cohort of **239K+ patients** using temporal preprocessing and feature engineering
- Designed and evaluated **15+** classification and regression experiments across **3 data modalities**: speech, ECG, and clinical records, using hyperparameter optimization and statistical performance analysis

Data Management Intern Jan. 2024 – May 2024
InfoStrux *Vancouver, BC*

- Optimized **25+** SQL queries in Snowflake, reducing execution time by up to **60%** and boosting performance of business intelligence dashboards
- Partnered with a senior data engineer to design Snowflake staging and curated layers for **3** datasets; wrote and tuned **30+** queries, improving data reusability and cutting time-to-insight by **40%**

Projects

Local LLM Agent System | *LLMs, RAG, AI Agents, MCP, REST APIs, Information Retrieval* GitHub

- Engineered a local multimodal RAG pipeline with **Qwen 3.5**, **Ollama**, and **Docling**, combining **Nomic embeddings**, hybrid retrieval, and **BGE reranking** for document search
- Indexed **100+** lecture documents for source-grounded Q&A with inline citations, improving top-5 retrieval recall from **64% to 86%** through BGE cross-encoder reranking
- Built an **MCP**-based Gmail agent on the same self-hosted LLM, automating **200+** outreach and follow-up emails for student-club recruiting and partnership outreach

Event Lottery System | *Java, Kotlin, Firebase, Gemini API, Google Maps* GitHub

- Collaborated in a **seven-person Agile team** to build a Firestore-backed Android platform supporting **3 user roles** and end-to-end lottery workflows, including waitlists, randomized selection, invitations, and automatic redraws
- Integrated QR-code registration, geolocation, and real-time notifications with a serverless **Gemini-powered classification pipeline** that mapped user-generated event tags across **21 validated categories**

Technical Skills

Languages: Java, Python, Go, C, C++, JavaScript, TypeScript, SQL
ML / AI: Machine Learning, Deep Learning, NLP, Computer Vision, LLMs, RAG, AI Agents, Time Series, Reinforcement Learning, Feature Engineering, Model Evaluation
Frameworks: PyTorch, TensorFlow, FastAPI, scikit-learn, Hugging Face, OpenCV, XGBoost, LightGBM, CatBoost, Django
Data & Infrastructure: PostgreSQL, MySQL, Snowflake, Firebase, Docker, Kubernetes, Redis, Linux, AWS S3, Git
AI Systems: MCP, REST APIs, Vector Search, Embeddings, Semantic Search, Local LLM Inference